

🏠 Kubernetes Lab Challenge: Logging & Monitoring Setup on AWS EKS

🎯 Objective

Set up **centralized logging and monitoring** for workloads running in AWS EKS using **CloudWatch** and **Prometheus/Grafana**.

🗺️ Challenge Overview

As part of a production-grade deployment, observability is key. Your task is to:

- Capture logs from containers and send them to **CloudWatch Logs**
- Monitor metrics using **Prometheus** and **Grafana**
- Use the AWS-managed Prometheus & Grafana services (AMP & AMG)

🧰 Tools & Services

- Fluent Bit or CloudWatch Agent
- Amazon CloudWatch Logs
- Amazon Managed Prometheus (AMP)
- Amazon Managed Grafana (AMG)
- EKS IRSA for secure access

🛠️ Tasks

🟢 Step 1: Enable CloudWatch Logs

- Configure Fluent Bit as a DaemonSet on your EKS cluster
- Route logs to CloudWatch Logs under a log group like `/eks/observability-demo``
- Use IRSA or access key with `logs:PutLogEvents`` permission

📖 References:

- <https://docs.aws.amazon.com/eks/latest/userguide/fargate-logging.html>
- <https://github.com/aws-samples/amazon-cloudwatch-container-insights>

🟢 Step 2: Setup Prometheus Metrics Collection

- Deploy the **AMP collector (Prometheus agent)** to scrape metrics from your workloads
- Ensure it has IRSA permissions to write to AMP workspace

📖 References:

- <https://docs.aws.amazon.com/prometheus/latest/userguide/AMP-onboard-create-workspace.html>
- <https://docs.aws.amazon.com/prometheus/latest/userguide/set-up-ingest.html>

🟢 Step 3: Visualize Metrics in Amazon Managed Grafana

- Create a workspace in **Amazon Managed Grafana**
- Connect it to your AMP workspace
- Import dashboards for CPU/Memory/Pod metrics

📖 References:

- <https://docs.aws.amazon.com/grafana/latest/userguide/Amazon-Managed-Grafana.html>

🟢 Step 4: Validate Logging and Monitoring

- Deploy a sample app (e.g., nginx or busybox)
- View:
 - Pod logs in CloudWatch Logs
 - Pod-level metrics in Grafana

Use commands like:

```
```bash
kubectl logs <pod-name>
kubectl port-forward svc/grafana 3000:3000
```
```

✅ Completion Criteria

- You can view real-time logs in CloudWatch
- Grafana dashboards show metrics from your EKS cluster
- Prometheus targets are discovered and actively scraping metrics

🚀 Bonus Tasks (Optional)

- Setup alerting in AMP/Grafana
- Export custom application metrics using `prometheus\_client` (Python/Java)
- Deploy Loki and Grafana for open-source logging instead of CloudWatch

😬 Tips & Hints

- IAM Role for Service Accounts (IRSA) is preferred over access keys
- Use Helm charts where applicable for Fluent Bit, Prometheus, Grafana
- Work with namespaces to isolate monitoring for apps

📄 Deliverables

Be ready to demonstrate:

- Grafana metrics dashboard
- CloudWatch logs for your app
- `kubectl get all -n observability` for deployed monitoring stack